

AAI-540 ML Design Document Template

Team Info

Project Team: Group 6

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Publication Date: 20/05/26

Team Workflows

GitHub Project Link: https://github.com/kbhakti/USD_MLOps

Asana Board Link:

<https://app.asana.com/1/952672460738672/project/1214960120378301/list/1214961363388782>

Team Tracker Link:

<https://docs.google.com/document/d/1BbpAceBOrEgWvCV5bWQnhdzYmw25duXufCgTw1Bb05Y/edit?usp=sharing>

Project Scope

Project Background:

Educational institutions often face challenges in identifying students who may require additional academic support before their performance declines significantly. Traditional methods of evaluating student performance are usually reactive, where interventions occur only after poor results are observed. This project aims to address this problem by leveraging Machine Learning techniques to predict student academic performance based on historical and behavioral data such as attendance, study time, parental education, previous grades, and extracurricular activities. By providing early predictions, the system can help educators make data-driven decisions and offer timely support to students who may be at academic risk.

The primary objective of the model is to classify students into performance categories such as “High Performance” or “Low Performance” using historical student-related data. This project focuses on solving a supervised Machine Learning classification problem, where labeled data is used to train models that can predict future student outcomes. The project also serves as a practical implementation of an end-to-end Machine Learning workflow, including data preprocessing, exploratory data analysis, model training, evaluation, and basic model monitoring concepts within a MLOps-focused learning environment.

Technical Background:

The project will utilize publicly available student performance datasets sourced from Kaggle (<https://www.kaggle.com/datasets/larsen0966/student-performance-data-set>). This dataset includes information related to student demographics, academic history, attendance records, study habits, and other education-related attributes. Data preprocessing will involve handling missing values, encoding categorical features, normalizing numerical data, and splitting the dataset into training and testing subsets to ensure reliable model development and evaluation. Exploratory Data Analysis (EDA) will be conducted to uncover patterns, analyze feature correlations, and understand the distribution of variables that may impact student performance. Features such as study time, attendance percentage, previous academic grades, parental education level, and participation in extracurricular activities are expected to play a significant role in predicting student outcomes.

The performance of the Machine Learning models will be evaluated using standard classification metrics, including Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. These metrics will help assess how effectively the models classify students into performance categories. In addition, the project will incorporate basic model monitoring practices, such as tracking model accuracy over time, identifying performance degradation, and detecting potential data drift when new data becomes available.

The initial set of algorithms considered for implementation includes Logistic Regression, Decision Tree, and Random Forest Classifier. Among these, Random Forest is expected to provide superior performance due to its ability to manage feature interactions effectively, reduce overfitting, and improve overall prediction accuracy.

Goals vs Non-Goals:

Goals

- Build a supervised Machine Learning classification model to predict student academic performance categories such as “High Performance” and “Low Performance.”
- Perform comprehensive data preprocessing, including handling missing values, encoding categorical variables, feature scaling, and dataset splitting to prepare high-quality training data.
- Conduct Exploratory Data Analysis (EDA) to identify trends, feature correlations, distributions, and factors that significantly influence student performance outcomes.
- Compare and evaluate multiple Machine Learning algorithms such as Logistic Regression, Decision Tree, and Random Forest Classifier using performance metrics including Accuracy, Precision, Recall, and F1-Score.
- Understand the importance and impact of features such as attendance, study time, previous grades, parental education, and extracurricular activities on model predictions.

- Demonstrate the complete end-to-end Machine Learning workflow, including data ingestion, preprocessing, model training, evaluation, and basic monitoring concepts aligned with MLOps practices.
- Explore basic model monitoring techniques by tracking prediction performance over time and identifying potential model degradation or data drift when new data is introduced.
- Develop a practical and educational project that showcases fundamental Machine Learning and MLOps concepts in a simple and understandable manner.

Non-Goals

- Building a production-grade educational analytics platform intended for deployment across schools or universities.
- Developing real-time student monitoring systems or live prediction pipelines integrated with classroom applications.
- Implementing Deep Learning architectures, advanced neural networks, or large-scale AI models, as the project focuses on foundational Machine Learning techniques.
- Integrating the solution with external educational management systems, APIs, or institutional databases.
- Designing cloud-native infrastructure, distributed systems, or scalable enterprise-level deployment pipelines.
- Creating a fully automated MLOps CI/CD pipeline with advanced orchestration, Kubernetes deployment, or automated retraining workflows.
- Collecting sensitive personal student information or building a system for high-stakes academic decision-making.
- Achieving state-of-the-art research-level accuracy, as the primary focus of the project is learning and demonstrating the Machine Learning lifecycle rather than maximizing production performance.

Solution Overview

The proposed solution is an end-to-end Machine Learning system designed to predict student academic performance using historical educational and behavioral data. The system will begin with data collection from. The collected data will be stored locally and processed through a data preprocessing pipeline that handles missing values, categorical feature encoding, normalization, and dataset splitting. Exploratory Data Analysis (EDA) and feature engineering will then be performed to identify the most influential factors affecting student performance, such as attendance, study time, previous grades, and parental education level.

After preprocessing, multiple supervised Machine Learning classification models, including Logistic Regression, Decision Tree, and Random Forest Classifier, will be trained and evaluated. Model performance will be compared using metrics such as Accuracy, Precision,

Recall, F1-Score, and Confusion Matrix. The best-performing model will be selected for deployment as a simple prediction service or local application prototype. Prior to releasing a new model version, testing will include validating model accuracy, checking data quality, ensuring preprocessing consistency, and evaluating performance on unseen test data. Basic monitoring practices will also be implemented by tracking prediction accuracy over time, identifying performance degradation, and detecting potential data drift when new data is introduced.

Public Datasets (Kaggle) -> Data Storage (CSV/Local Files) -> Data Processing -> EDA -> Model Training -> Model Evaluation -> Model Deployment -> Monitoring and Testing

Data Sources:

What is your data source?

- The project will use publicly available student performance datasets from Kaggle. This dataset contains student demographic information, academic history, attendance records, study habits, parental education details, and extracurricular activity participation.

What is your data volume?

- The dataset size is ~93 KB. Since this is an academic MLOps learning project, the data volume is moderate and manageable on a local machine.

Why did you select this data set?

- The dataset was selected because it is publicly accessible, well-structured, and commonly used for educational Machine Learning projects. It contains multiple academic and behavioral factors that are suitable for demonstrating supervised classification techniques and end-to-end ML workflows.

Any risks (bias, sensitive features, etc)?

- Potential risks include dataset bias related to demographic or socioeconomic factors such as parental education and family background. Some attributes may indirectly influence predictions in ways that could create fairness concerns. The project will avoid making high-stakes academic decisions and will use the model only for educational and learning purposes.

Data Engineering:

How will you store this data?

- The data will be stored locally in CSV format during development and experimentation. Processed datasets and trained model artifacts may also be stored in local directories for reproducibility.

What data pre-processing do you need to do before you feed it into your ML system?

- The preprocessing steps will include:
 - o Handling missing values
 - o Removing duplicate records if present
 - o Encoding categorical variables
 - o Feature normalization/scaling
 - o Splitting data into training, validation, and testing sets
 - o Basic outlier analysis and handling if required

Training Data:

How will you split your data into training, test and validation?

The dataset will be divided into:

- Training Set: 70%
- Validation Set: 15%
- Test Set: 15%

The training set will be used for model learning, the validation set for hyperparameter tuning and model comparison, and the test set for final evaluation.

Will you use any data labeling techniques?

- No additional labeling techniques will be required because the selected datasets are already labeled with student performance outcomes or grade categories.

Feature Engineering:

Which fields from your data will you use or exclude?

- Fields such as age, gender and parental education etc. Fields with minimal relevance, excessive missing values, or direct identifiers (if present) will be excluded.

Which fields will be combined or bucketed?

- Certain numerical values such as grades or attendance percentages may be grouped into categories such as “High,” “Medium,” and “Low.” Similar study-related features may also be combined into aggregated performance indicators.

What other data transformations will you apply to your data?

- Additional transformations may include:

- One-hot encoding for categorical features
- Feature scaling using normalization or standardization
- Correlation analysis for feature selection
- Handling class imbalance if necessary

Model Training & Evaluation:

How will you train your model?

- The model will be trained using supervised learning techniques on historical student data after preprocessing and feature engineering. Multiple algorithms will be compared to identify the best-performing model.

What algorithm will you use?

- The primary algorithms considered are:
 - Logistic Regression
 - Decision Tree
 - Random Forest Classifier

Random Forest is expected to perform best because of its ability to manage feature interactions and reduce overfitting.

What parameters will you use?

- Initial parameters may include:
 - Number of estimators for Random Forest
 - Maximum tree depth
 - Minimum samples split
 - Random seed for reproducibility

Hyperparameter tuning may be performed using Grid Search or Random Search.

How will you evaluate your model?

- The model will be evaluated using:
 - Accuracy
 - Precision
 - Recall
 - F1-Score
 - Confusion Matrix

These metrics will help measure classification performance and prediction reliability.

Model Deployment:

What instance size will you use?

- Since this is a lightweight academic project, the model will run on a local machine or small compute environment without requiring large cloud infrastructure.

Will your model function as a batch or real time model? Why?

- The model will function as a batch prediction system because predictions will be generated on static datasets rather than continuous real-time student data streams. This keeps the implementation simple and aligned with the project scope.

Model Monitoring:

How will you monitor your model?

- The project will monitor:
 - o Prediction accuracy over time
 - o Changes in evaluation metrics
 - o Model performance degradation

How will you monitor your infrastructure?

- Infrastructure monitoring will be minimal because the solution will primarily run locally. Basic logging and runtime observation will be used during experimentation.

How will you monitor your data?

- The project will monitor:
 - o Missing value patterns
 - o Data quality issues
 - o Feature distribution changes
 - o Potential data drifts when new data is introduced

Model CI/CD:

What checkpoints will your CI/CD pipeline contain?

- Potential checkpoints include:
 - o Data validation
 - o Model training completion
 - o Model evaluation threshold checks
 - o Model artifact generation
 - o Deployment approval stage

What tests will your CI/CD pipeline contain?

- The pipeline may include:
 - o Data quality validation tests
 - o Unit tests for preprocessing scripts
 - o Model performance threshold validation
 - o Integration testing for prediction workflow

Security Checklist, Privacy and Other Risks:

Will this store or process Personal Health Information (PHI)?

- No.

Will this store or process Personal Identifiable Information (PII)?

- No personally identifiable information will intentionally be stored or processed. Public datasets used are anonymized for educational purposes.

Will user behavior be tracked and stored?

- No.

Will this store or process credit card information?

- No.

If you answered yes to any of the above questions, please justify.

- Not applicable.

What S3 buckets will this application read from or write to?

- This project does not currently use AWS S3 buckets because it is designed as a local academic implementation.

What data bias should be considered?

- Bias may exist due to demographic, socioeconomic, or educational inequalities present in the original dataset. Certain features may unintentionally influence predictions unfairly.

Will your model have potential for bias along sensitive features (race, ethnicity, gender, age, religion, disability, sexual orientation, or other personal attributes)

- Yes, there is potential for indirect bias related to parental education, socioeconomic background, age, or other demographic attributes. These risks will be acknowledged during evaluation and analysis.

Are there any ethical concerns with the data or business problems that should be addressed?

- Yes. The model should not be used as the sole basis for academic judgment or disciplinary action. Predictions should only support educators in identifying students who may benefit from additional assistance. Fairness, transparency, and responsible AI usage should be considered throughout the project.

Future Enhancements:

Develop a Real-Time Prediction and Monitoring System

- The current implementation focuses on batch-based predictions using static datasets. As a future enhancement, the system could be upgraded into a real-time student performance monitoring platform capable of processing live academic data, generating continuous predictions, and sending early alerts for students who may require academic support.

Integrate Cloud-Based MLOps Infrastructure

- The project could be enhanced by deploying the complete ML pipeline on cloud platforms such as AWS, Azure, or Google Cloud. This would include automated CI/CD pipelines, scalable model deployment, experiment tracking, automated retraining workflows, and cloud-based monitoring for model performance and data drift.

Improve Explainability and Responsible AI Features

- Future versions of the system could incorporate explainable AI techniques such as SHAP or LIME to provide transparency into model predictions. This would help educators understand which features most influence student performance predictions and support more responsible and interpretable AI usage.

Expand Dataset Diversity and Feature Coverage

- Additional datasets from multiple educational institutions or regions could be incorporated to improve model generalization and reduce dataset bias. More features such as student engagement metrics, online learning activity, and behavioral trends could also be included to strengthen predictive capabilities.